

Bernstein Energy & Power: The rise of 'Electro-states'...

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By Nikhil Nigania and Uma Menon,

The fundamental point we are trying to make in this note is that 'Electricification' is not a cyclical but a structural shift. Among countries, eventual beneficiaries would be - electrification equipment suppliers (China), highly electrified economies (we are hoping for acceleration in India post West Asia war), or even electrification raw material exporters (e.g. Australia, Chile). Within countries, our view continues to be a) [Transmission](#) is the heart of this shift and the single biggest constraint, b) Role of [conventional generation](#) is under-stated in the transition phase, c) Similarly, the role of [storage](#) is likely to be even bigger than industry estimates.

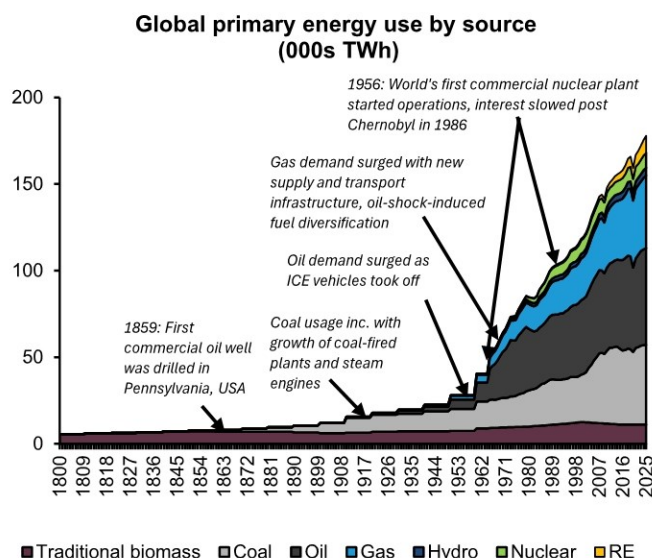
A) ELECTRIFICATION A STRUCTURAL TREND...NOW ACCELERATED BY DC, EV AND THE WAR

Oil-Gas gained share from coal in the last century as they were cleaner, more energy dense, easier to control, and to handle/transport, leading to the rise of petro-states, with Oil discovery in Saudi in 1938, gas discovery in Qatar in 1971 etc. Now there is a shift in share from usage of Oil-Gas products as the final form of energy consumption, to electricity as the final form, and we see a good case for the rise of electro-states ('a country which uses electricity as its primary source of energy'). Oil drove the defining technologies of the 20th century - cars, airplanes etc., Electricity is driving it in the 21st century in Data-centers, EVs and in cooling a warmer world.

Globally, electricity's share of energy consumption is expected to rise from ~23% today to ~30% by 2050, with China leading as expected. A big part of China's rise in the last few decades has been electricity - with its share in final energy consumption rising from 17% in 2010 to ~30% now, and expected to reach 46% by 2050. China today accounts for ~50% of global solar & wind capacity and 56% of global coal based capacity, adding more generation capacity in ~2 years than the US has in total. Whether it's short-distance transport or long distance the whole economy is incrementally designed on electricity. Given DCs, EVs and the

war, we expect other countries (especially Oil-Gas importers) to accelerate this shift as well, mimicking China to the extent possible.

EXHIBIT 1: History of primary energy use by source

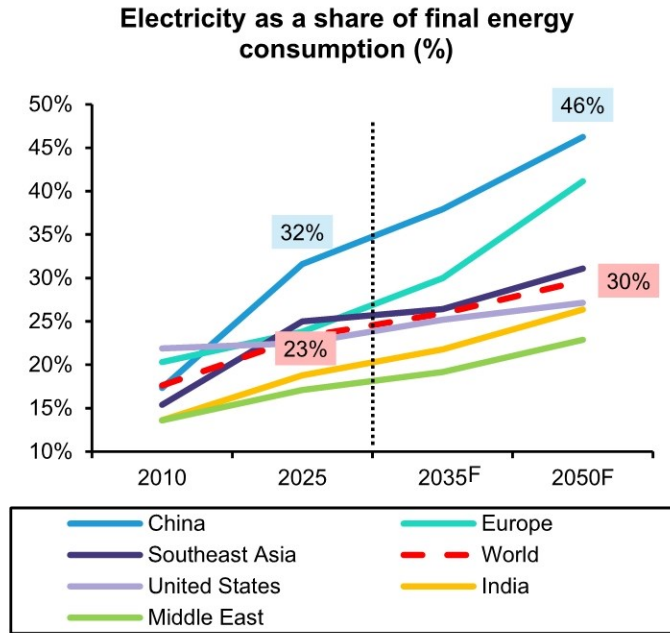


Renewables includes solar, wind, Bio-fuels and other renewables
 Source: Our World in Data, Bernstein analysis

In developing countries the push is more from industries, buildings and appliance ownership, in developed markets it is more DCs and EVs driven.

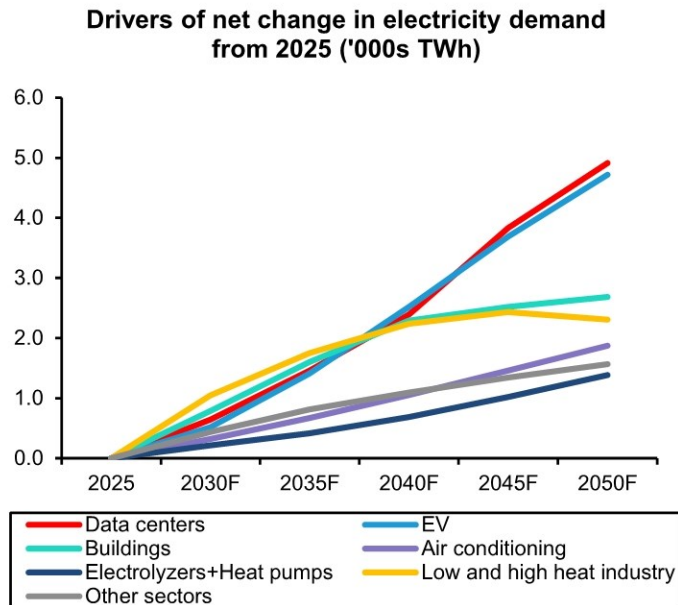
(Continued...)

EXHIBIT 2: Electricity as a share of final energy consumption expected to rise globally to 30% from 23% today



Source: IEA, Bernstein analysis

EXHIBIT 3: Key drivers of global electricity demand



Source: BNEF, Bernstein analysis

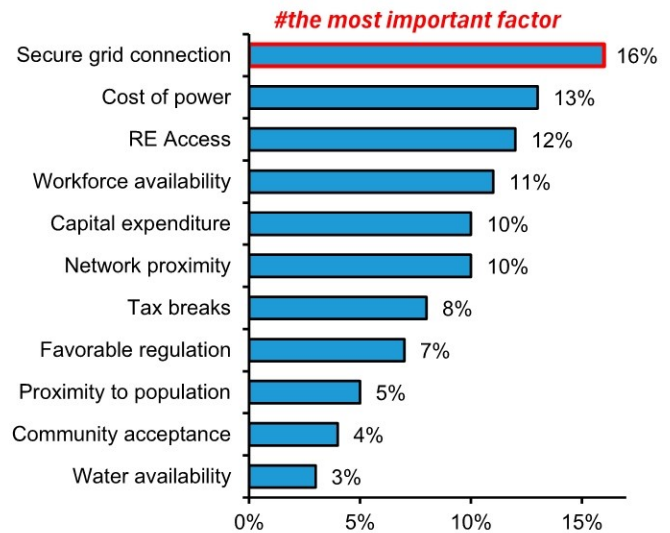
B) WHY IS ELECTRICITY AN ADVANTAGE?

I. DATA CENTERS - ELECTRICITY THE BIGGEST CONSTRAINT

The main parameters in the decision for location of DCs are electricity availability (and cost), land, fiber network, tax breaks and existing DC ecosystem in the region. Hence, by nature US wins given its large established DC set-up, fiber network and customer base - but now gradually grid-land (not in my backyard) constraints in US hot-spots are increasing attraction of markets beyond US as well. Highly electrified countries investing strongly in their grid build-out stand to eventually gain a share of the DC build-out. Here, countries like China-India where an extra high voltage line can be built in 2-3 years, are much better placed than US-Europe where it can take 7-8 years for such a line.

EXHIBIT 4: Factors for DC site selection: Top 3 are Electricity - cheap, clean and fast grid connectivity

Key factors for data center site selection

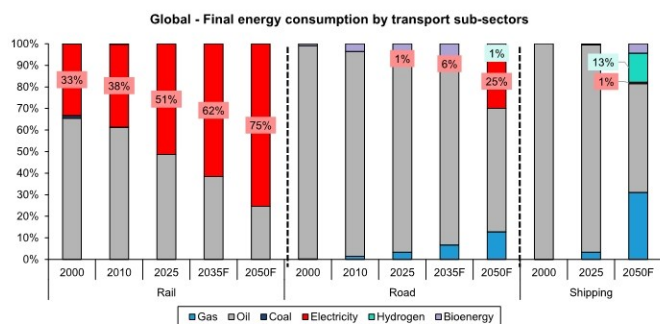


Source: BNEF, Bernstein analysis

II. TRANSPORT - INCREASINGLY ELECTRIFIED

If we look at transport - long distance - China is the best example with ~50,000 km of electricity powered high speed rail network reducing excessive dependence on air travel. On short distance the rising penetration of EVs is well understood.

EXHIBIT 5: For long distance travel, however, electricity share in total energy demand to be substantial only for railways and road transport by 2050

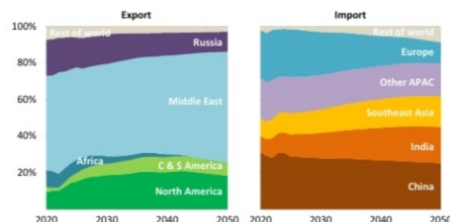


Source: Bloomberg, Bernstein analysis

III. ENERGY SECURITY - FOR OIL-GAS IMPORTERS ELECTRICITY IS THE ANSWER

If we look at the Oil-Gas importers, China-Japan-India make up the biggest ones. Both China and India have one of the lowest cost of solar and coal based electricity generation globally. This makes electrification a big tool for energy security, and hence with the war - grid and power generation build-out has only accelerated.

EXHIBIT 6: Oil importers and exporters- The biggest importers are best positioned for electrification



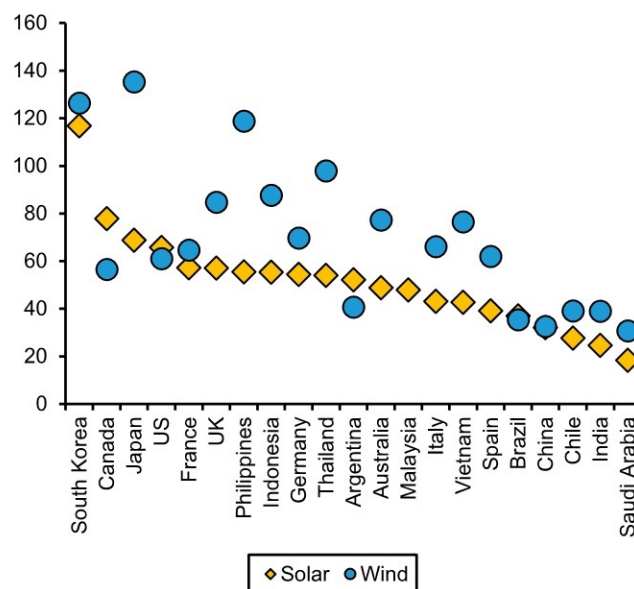
Source: IEA

C) WHICH COUNTRIES ARE BETTER POSITIONED?

On cost of generating electricity, India-China and most emerging markets are well positioned - especially on renewable power. On gas-based generation, most don't look great given their import dependence, but on coal-based generation as well - these big oil-gas importers are very well positioned. While China has a big head-start in the electrification shift, we expect India and other EMs to accelerate as well.

EXHIBIT 7: LCOE for solar and wind power by country

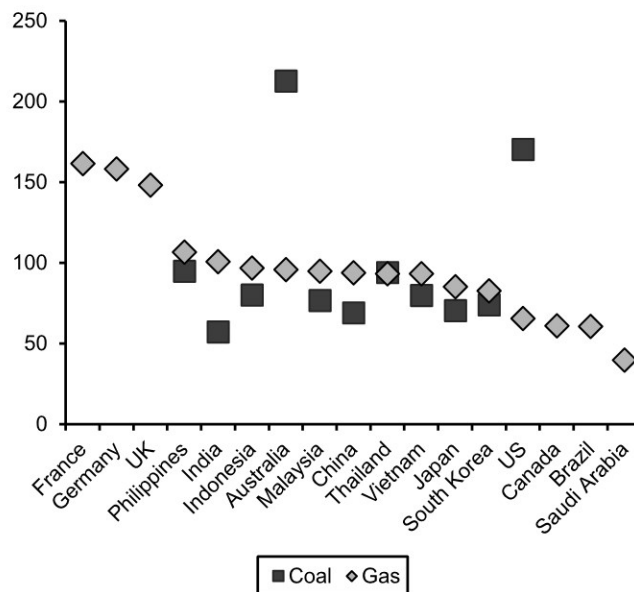
2025 Solar and wind LCOE (\$/MWh)



Source: Bloomberg, Bernstein analysis

EXHIBIT 8: LCOE for coal and gas based power by country

2025 Coal and gas (CCGT) LCOE (\$/MWh)

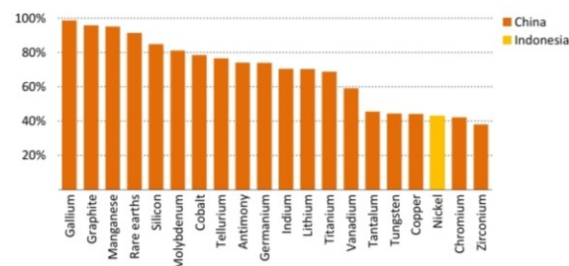


Source: Bloomberg, Bernstein analysis

One risk in the electrification story though is the excessive dependence on one nation, vs. the multiple nation option in the Oil import-export market. One nation has such a strong hold across the supply value chain linked to electrification - from raw-material refining, to equipment manufacturing that it gives it extreme power

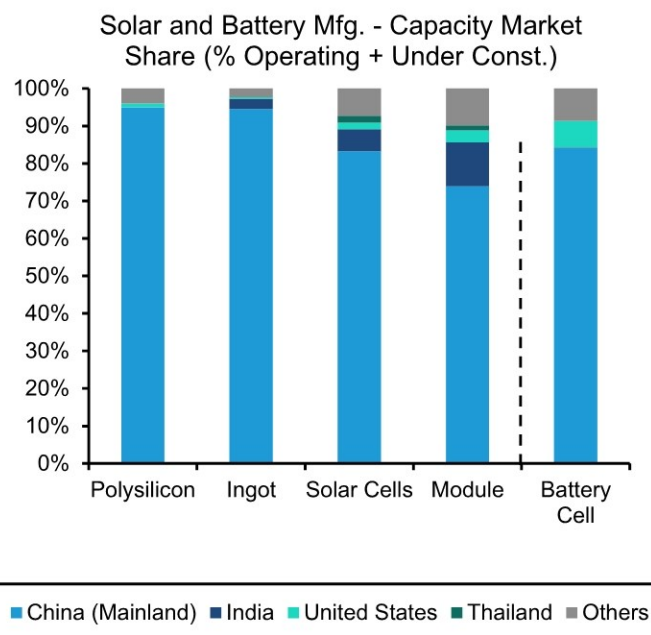
over countries looking to build themselves as 'electro-states'. However, other potential electro-states like India are becoming increasingly cognizant of a local supply-chain and are building the same.

EXHIBIT 9: China controls refining for most of the strategic energy mineral



Source: IEA

EXHIBIT 10: Equipment value chain: Completely dominated by China, but other countries looking to promote local manufacturing



Source: Bloomberg, Bernstein analysis

BERNSTEIN TICKER TABLE

			30 Sep 2026		TTM	Reported EPS				Reported P/E (x)		
Ticker	Rating	Cur	Closing	Price	Rel.							
			Price	Target	Perf.	Cur	2026A	2027E	2028E	2026A	2027E	2028E
NTPC.IN (NTPC Ltd)	O	INR	322.75	450.00	(29.4)%	INR	27.90	27.06	28.42	11.6	11.9	11.4
ASIAX			1,944.04									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended
Source: Bloomberg, Bernstein estimates and analysis.

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We value NTPC using DCF method to arrive at our INR450 price target. On the conventional business, we forecast the financials till 2070 asset-wise - assuming no further thermal/nuclear capacity addition. For renewables, we add NTPC Green's valuation with a 30% holdco discount. We have used a weighted average cost of capital (WACC) of 8%.

RISKS**NTPC Ltd**

Downside:

1. Power demand in India growing slower than our anticipation of 5-6% CAGR
2. Uncertainty on ordering of thermal plants beyond the under-construction capacity, given competition from private sector
3. Further disappointment in renewable capacity addition by its subsidiary
4. Nuclear - No traction in nuclear project award

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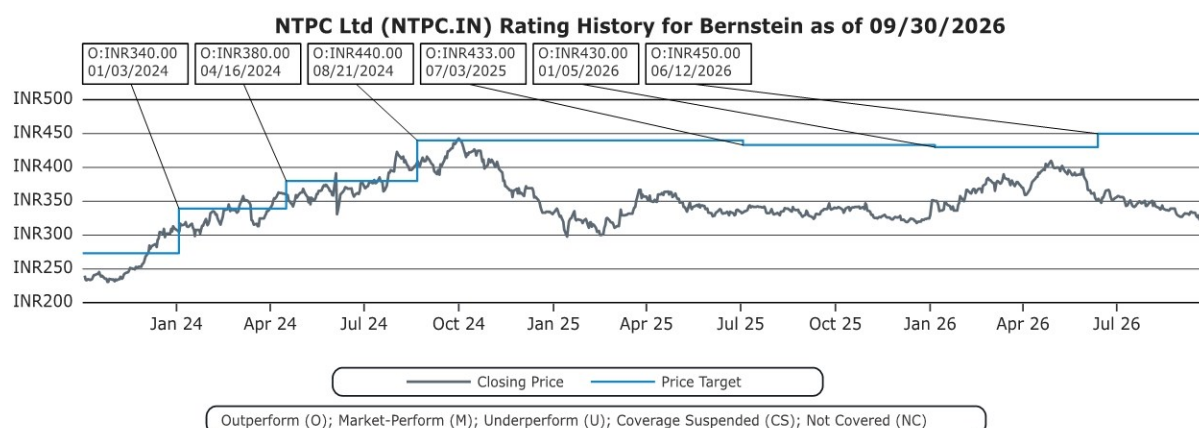
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Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.6%	16.1%
Underperform	SELL	12.1%	13.6%

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